Contributions on Schedule-Aware Bundle Routing

1 Background

Delay-Tolerant Networking (DTN) enables nodes to operate using a **store-carry-forward** strategy, addressing the long delays inherent to interplanetary distances and the extended disruptions caused by orbital dynamics and planetary rotations. Although such disruptions can be significant, they are **highly predictable**, allowing operators to construct a **contact plan**—a schedule that lists the future **contacts**, i.e., time intervals during which connectivity exists between pairs of nodes.

The **Schedule-Aware Bundle Routing (SABR)** standard, defined by the **Consultative Committee for Space Data Systems (CCSDS)**, specifies the recommended approach for routing in deterministic and scheduled time-varying topologies.

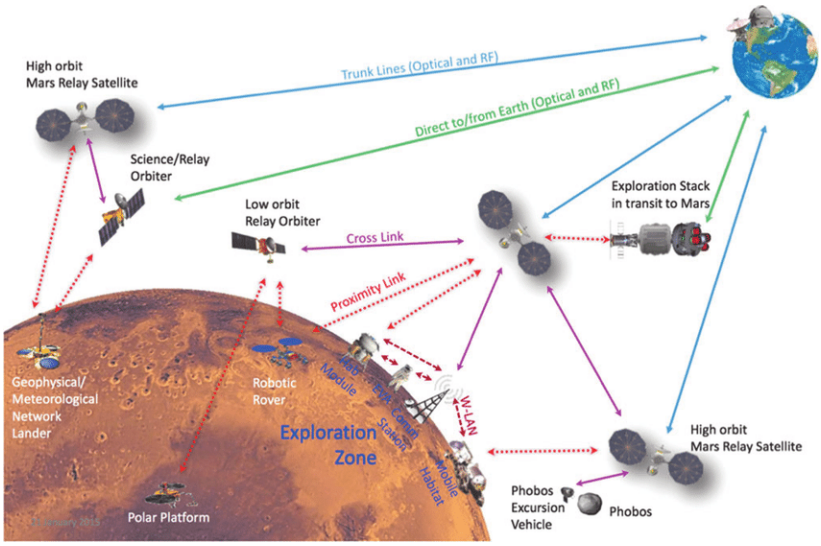

Figure 1: NASA/SCaN's Next Generation Mars telecommunication architecture to enable long-term human exploration. Source: Reinhart et al. (2017).

2 Task examples

The **Adaptive Library for Schedule-Aware Bundle Routing (A-SABR)** developed at **LIRMM**, provides a framework to implement various SABR variants proposed by the community. The library is open source in a "beta" release, written in **Rust**, and additional efforts are required to extend its capabilities, portability, and quality. Depending on the intership or module context, **one or more tasks** shall be completed:

- Implementation of features such as contact segmentation with priority support, inter-regional support, etc.
- Port to no-std for microcontroller support
- Integration in bundle protocol implementations like µD3TN, Hardy, ION, HDTN, or unibo-bp, etc.
- Implementation of a configuration tab in DTChat (Rust/egui)
- DevOps-related tasks such as unit/integration tests and code quality checks with GitHub Actions pipelines, containerization, etc.

3 Required skills

Interest in low-level programming with C, C++, or Rust. Interest in software engineering and microcontroller development. Further skills might be required depending on the selected tasks.

4 Some references

- **On Schedule-Aware Bundle Routing:** J. A. Fraire, O. De Jonckère, & S. Burleigh. Routing in the space internet: A contact graph routing tutorial. *Journal of Network and Computer Applications*, 174, 102884.
- **About the library:** O. De Jonckère, L. Ma, & J. A. Fraire. A-SABR: The Adaptive Library for Schedule-Aware Bundle Routing. *WiSEE 2025 - IEEE International Conference on Wireless for Space and Extreme Environments*, Oct 2025, Halifax, Canada.

5 Contact

The project will be supervised by Olivier De Jonckère. Interested students should send an e-mail to `olivier.de-jonckere@lirmm.fr` for additional information.